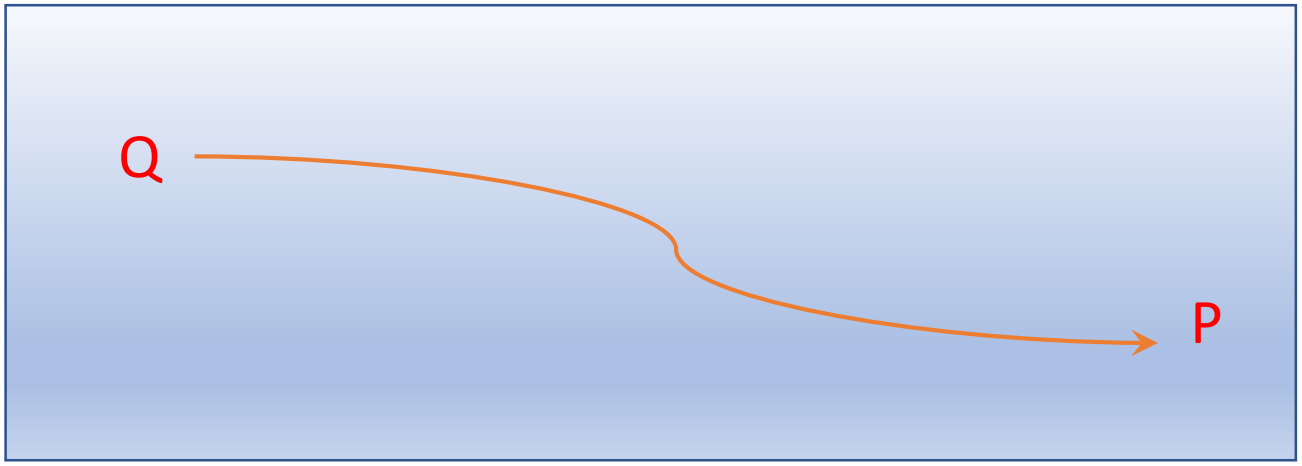
§3 费马原理

§3.1 光程

§3.2 费马原理的表述

§3.3 由费马原理推导几何光学三定律

§3.1 光程



光程：光在一定时间内在真空中所传播的距离

$$(QP) = \int_Q^P c dt = \int_Q^P c \frac{dl}{v} = \int_Q^P n dl$$

→ 光在介质中传播的距离dl与介质折射率n的乘积

在均匀介质体系中 $(QP) = nl$

§3.1 光程

光程： $$(QP) = \sum_i n_i \Delta l_i = n_1 \Delta l_1 + n_2 \Delta l_2 + n_3 \Delta l_3$$

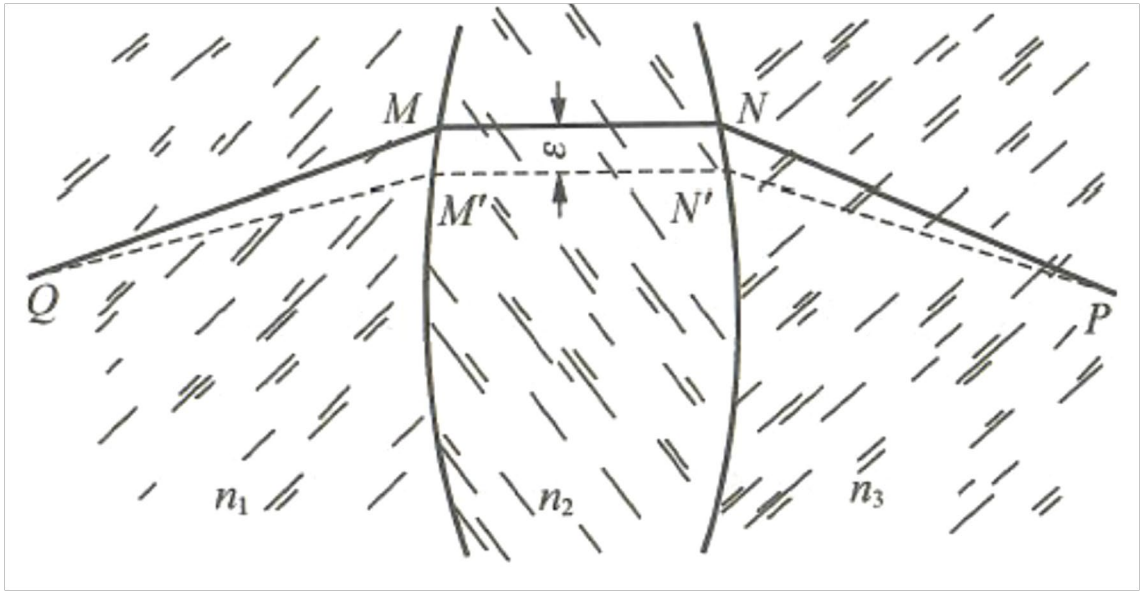
§3.2 费马原理的表述

QP两点间光线的实际路径，是光程(QP)（或者说所需的传播时间τ_{QP}）为平稳的路径。

$$\delta(QP) = \delta \int_Q^P n dl = 0 \qquad \text{或} \qquad \boldsymbol{\delta\tau_{QP} = 0}$$

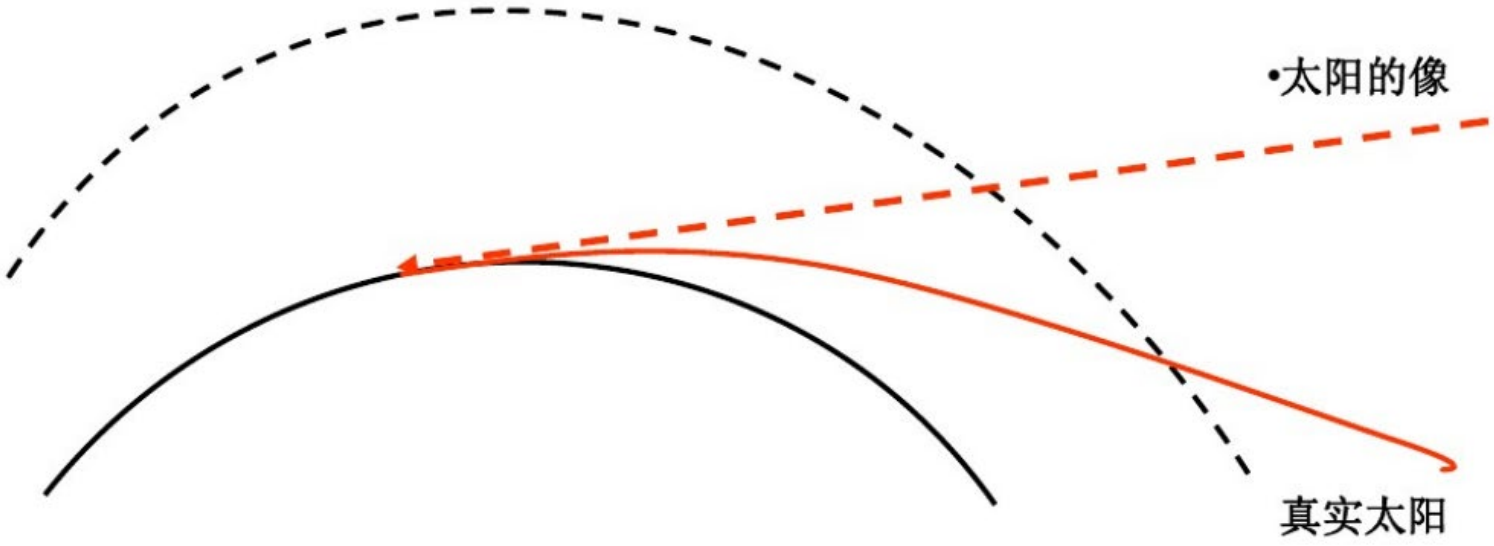
数学表述：在光线的实际路径上光程的变分为0.

极小
极大
常数
拐点
…



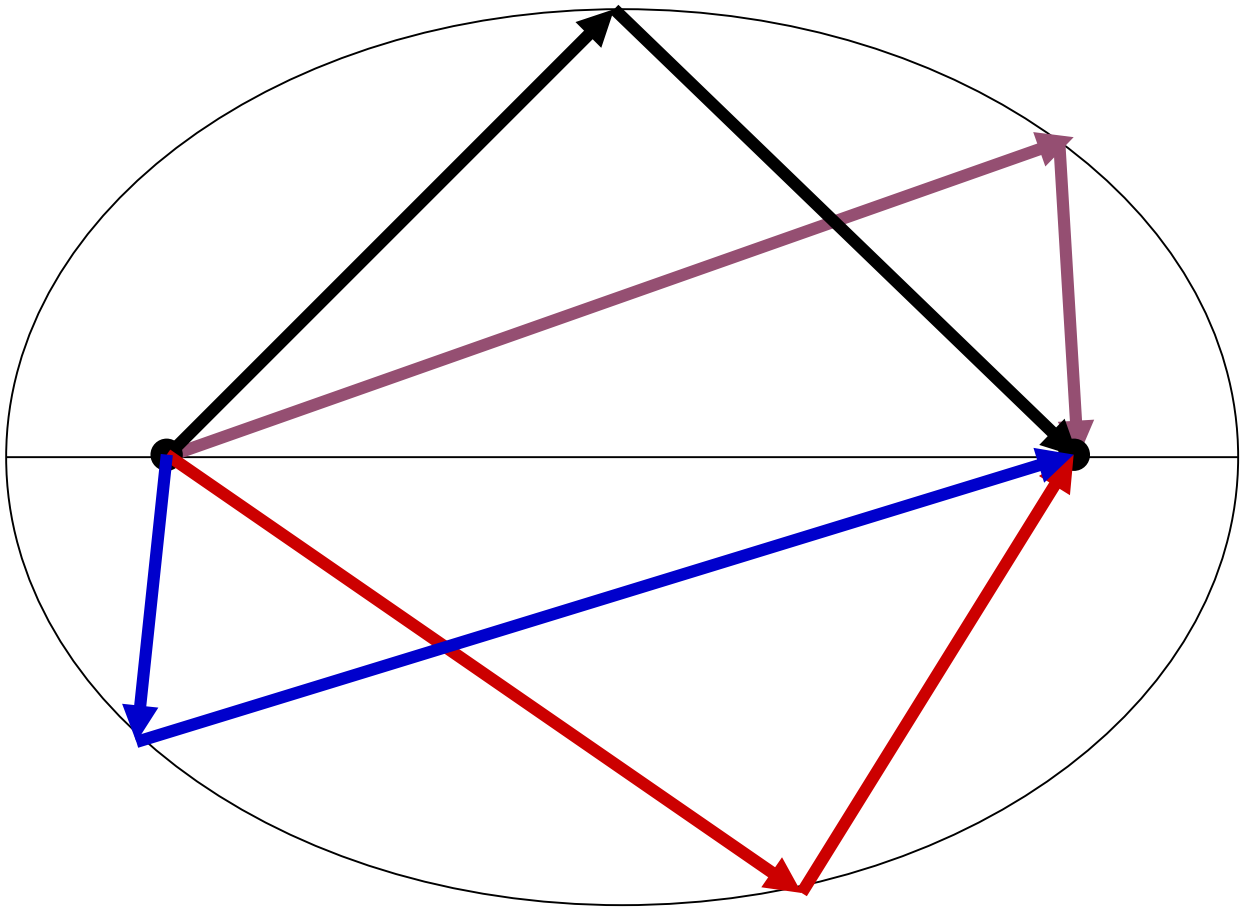
日出、日落

- 地球的大地高处稀薄，底部稠密，光在真空中传播要比空气中快，因而太阳光不沿地平线地平线进行。

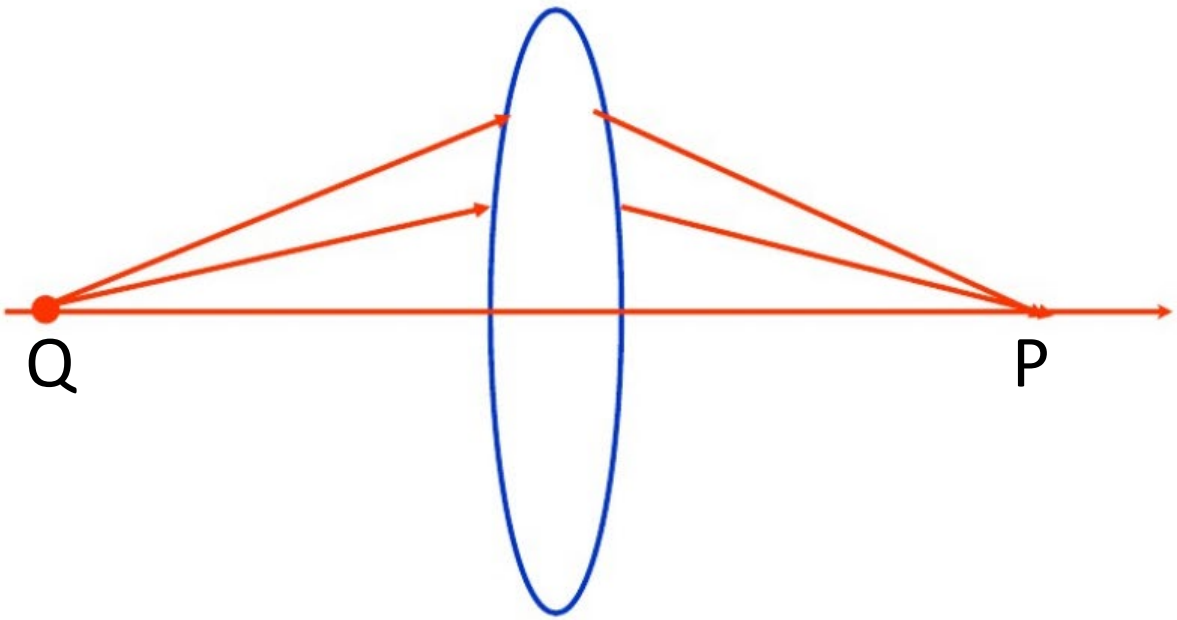


光程取常数的例子

椭球面内两焦点间光的路径，光程为恒定值

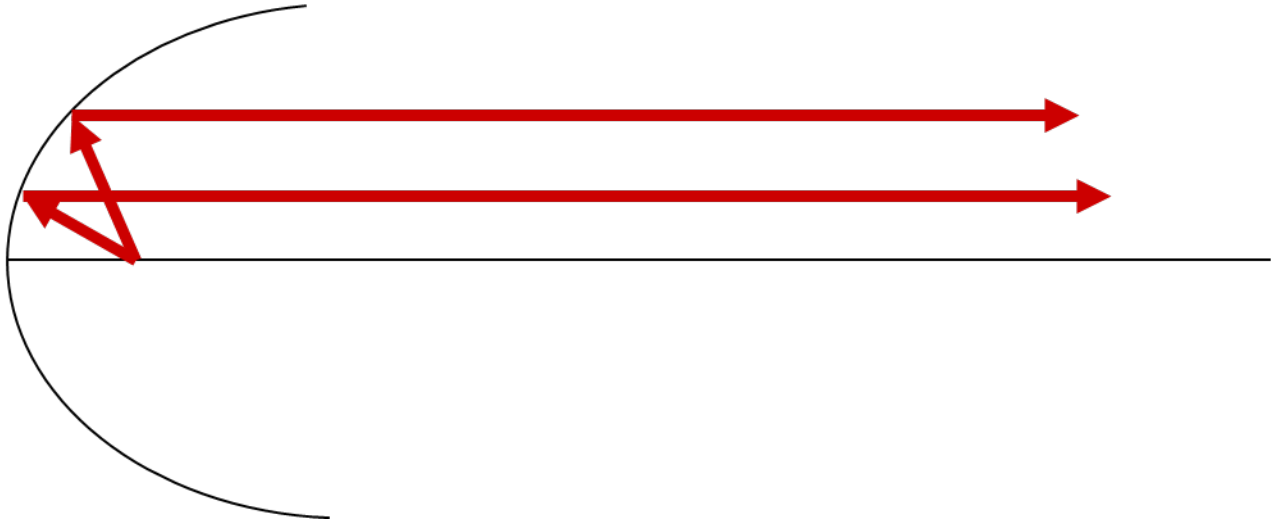


薄透镜成像的等光程性



光程取极大值的例子

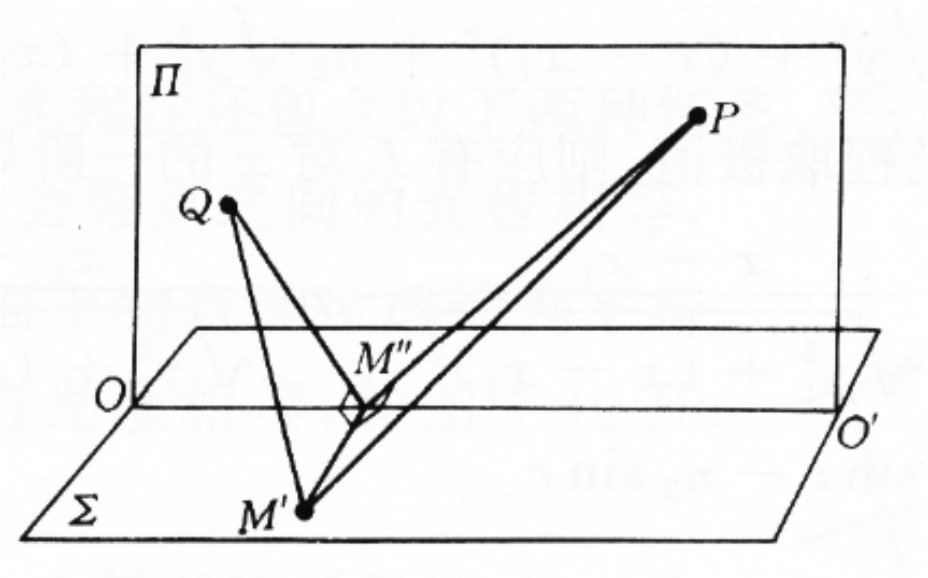
抛物面焦点发出的光，反射后变为平行光，汇聚在无穷远处，光程为极大值。



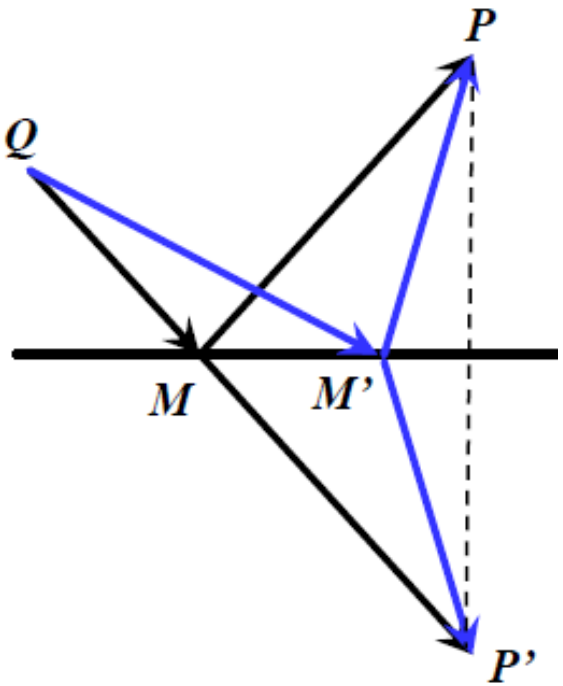
§3.3 由费马原理推导几何光学三定律

（1）光的直线传播定律：光在均匀介质中沿直线传播。

（2）反射定律



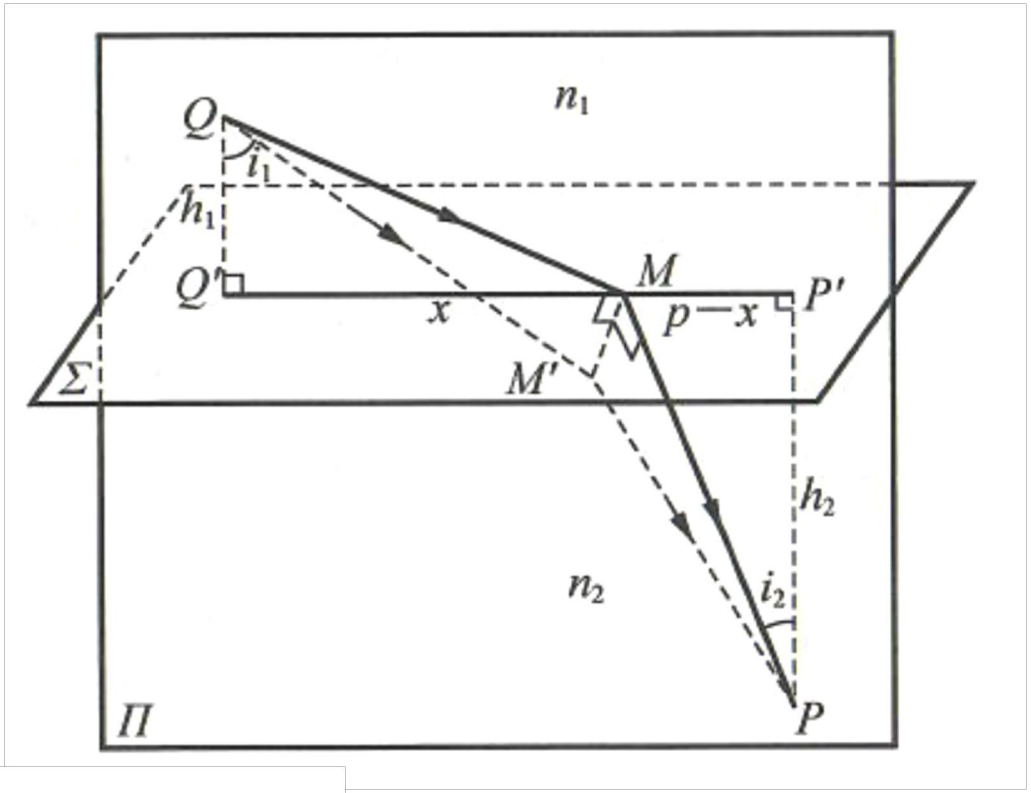
① 面内光程最短



② QMP光程最短

（2）折射定律

首先说明光程最短的路径在入射面内



$$(QMP)=n_1\,\overline{QM}+n_2\,\overline{MP}$$
$$=n_1\sqrt{h_1^2+x^2}+n_2\sqrt{h_2^2+(p-x)^2}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(QMP)=\frac{n_1x}{\sqrt{h_1^2+x^2}}-\frac{n_2(p-x)}{\sqrt{h_2^2+(p-x)^2}}$$
$$=n_1\sin i_1-n_2\sin i_2.$$

$$n_1 sin i_1 = n_2 sin i_2$$